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NEET · PHYSICS · CHAPTER-WISE PYQS

Current Electricity

NEET Physics · Previous Year Questions with Answer Key

116	36	2000 – 2026	English
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INSIDE THIS PDF

MCQ (Single Correct Answer)	116 questions
Answer key	at the end

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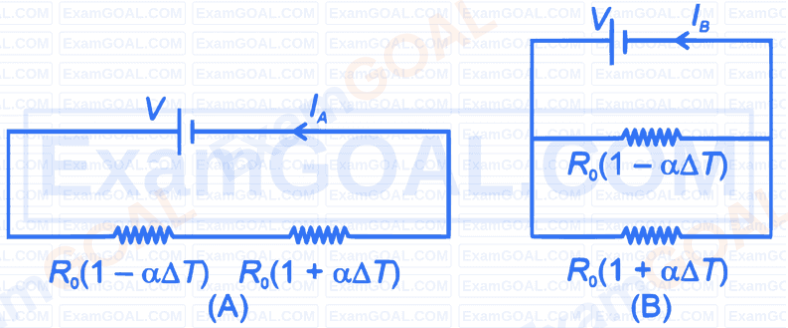
1. **Solve**
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2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

MCQ (Single Correct Answer) 116 questions

Q1. RE-NEET 2026

Open solution

Consider two circuits, (A) and (B), each having two resistors. One of them has a positive temperature coefficient of resistance, $+\alpha$, while the other one has a negative temperature coefficient, $-\alpha$, as shown in the figure. The currents through these circuits are denoted by I_A and I_B. At initial temperature, the resistance of the two resistors is R_0. As the temperature is increased, the correct option that describes the variation of current in these circuits is:



(A) Both I_A and I_B remain constant

(B) I_A remains constant while I_B increases

(C) I_A decreases while I_B increases

(D) I_A increases while I_B decreases

Q2. NEET 2026

Open solution

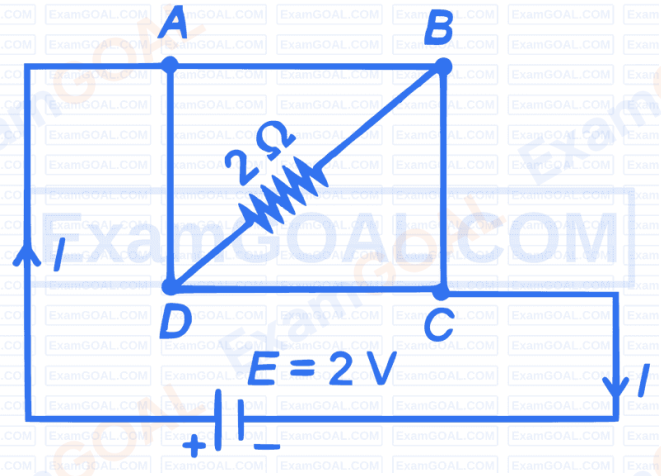
A resistor is connected to a battery of 12 V emf and internal resistance 2Ω. If the current in the circuit is 0.6 A , the terminal voltage of the battery is:

(A) 10 V (B) 1.2 V (C) 12 V (D) 10.8 V

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Q3. NEET 2026

A uniform metallic wire having resistance 4Ω is bent to form a square loop (ABCD) (see figure). A resistance of 2Ω is connected between points B and D and a battery of 2 V is connected across points A and C as shown in the figure. Now the value of current (I) is :



Open solution

(A) 2 A (B) 8 A (C) 4.5 A (D) 4 A

Q4. NEET 2026

A room heater is rated $400\ \text{W}, 220\ \text{V}$. If the supply voltage drops to 200 V , what will be the power consumed (approximately)?

(A) 200 W (B) 400 W (C) 331 W (D) 121 W

Open solution

Q5. NEET 2026

A galvanometer of resistance 100Ω gives full scale deflection for a current of 1 mA . It is converted into an ammeter of range $0 - 10\ \text{A}$. The shunt required is:

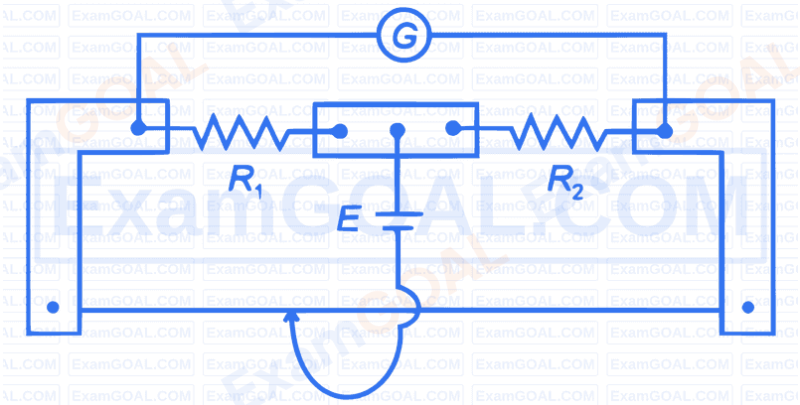
(A) 0.01Ω (B) 0.10Ω (C) 1.0Ω (D) 0.001Ω

Open solution

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Q6. NEET 2026

In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer:



Open solution

(A) Only the left-sided deflection

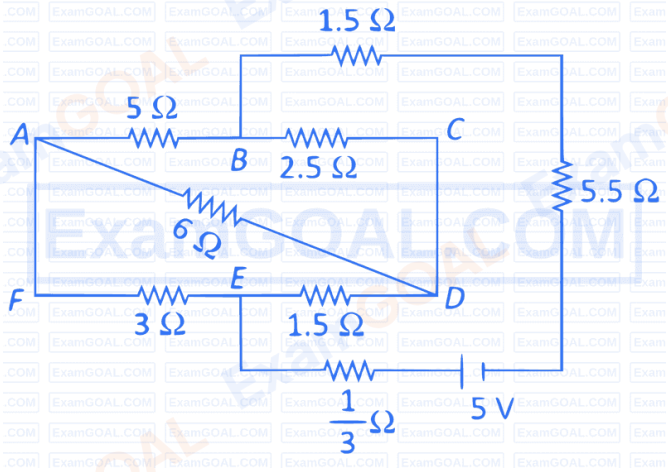
(B) There will be no deflection irrespective of the position of the jockey

(C) Only the right-sided deflection

(D) Both right-sided and left-sided deflection and at balance point, no deflection

Q7. NEET 2025

The current passing through the battery in the given circuit, is:



Open solution

(A) 2.5 A (B) 1.5 A (C) 2.0 A (D) 0.5 A

Q8. NEET 2025

A wire of resistance R is cut into 8 equal pieces. From these pieces two equivalent resistances are made by adding four of these together in parallel. Then these two sets are added in series. The net effective resistance of the combination is:

Open solution

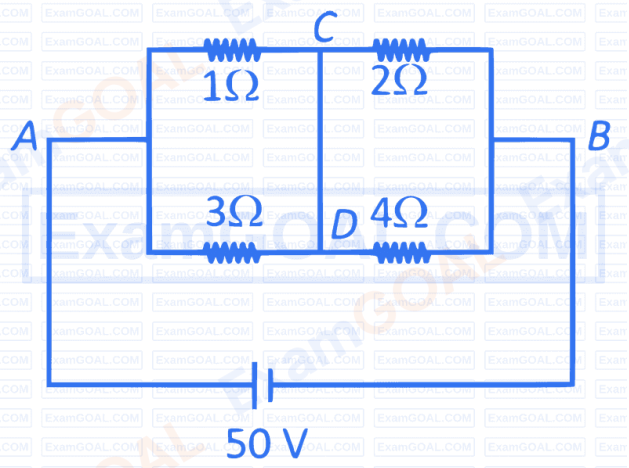
(A) $\frac{R}{16}$ (B) $\frac{R}{8}$ (C) $\frac{R}{64}$ (D) $\frac{R}{32}$

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Q9. NEET 2025

A constant voltage of 50 V is maintained between the points A and B of the circuit shown in the figure. The current through the branch CD of the circuit is:

Open solution



(A) 2.5 A (B) 3.0 A (C) 1.5 A (D) 2.0 A

Q10. NEET 2024 (Re-Examination)

A uniform wire of diameter d carries a current of $100\ \text{mA}$ when the mean drift velocity of electrons in the wire is v. For a wire of diameter $\frac{d}{2}$ of the same material to carry a current of $200\ \text{mA}$, the mean drift velocity of electrons in the wire is

Open solution

(A) $4v$ (B) $8v$ (C) v (D) $2v$

Q11. NEET 2024 (Re-Examination)

A uniform metal wire of length l has 10Ω resistance. Now this wire is stretched to a length $2l$ and then bent to form a perfect circle. The equivalent resistance across any arbitrary diameter of that circle is

Open solution

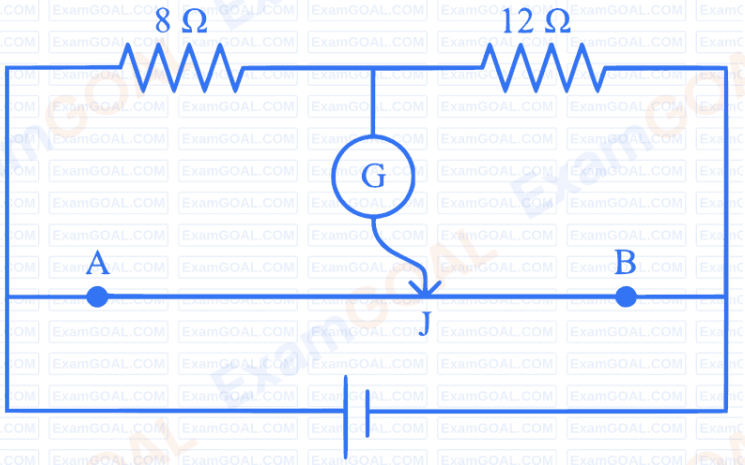
(A) 10Ω (B) 5Ω (C) 40Ω (D) 20Ω

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Q12. NEET 2024 (Re-Examination)

The given circuit shows a uniform straight wire AB of $40\ \text{cm}$ length fixed at both ends. In order to get zero reading in the galvanometer G, the free end of J is to be placed from B at:

Open solution

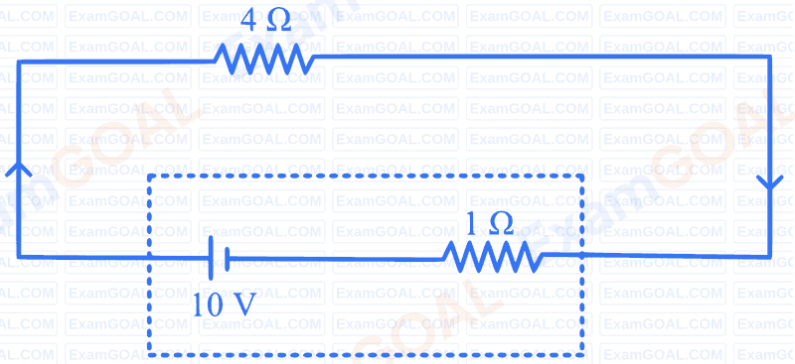


(A) 32 cm (B) 8 cm (C) 16 cm (D) 24 cm

Q13. NEET 2024

The terminal voltage of the battery, whose emf is $10\ \text{V}$ and internal resistance 1Ω, when connected through an external resistance of 4Ω as shown in the figure is:

Open solution



(A) 4 V (B) 6 V (C) 8 V (D) 10 V

Q14. NEET 2024

A wire of length 'l' and resistance 100Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is:

Open solution

(A) 26Ω (B) 52Ω (C) 55Ω (D) 60Ω

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Q15. NEET 2024

Two heaters A and B have power rating of $1\,\text{kW}$ and $2\,\text{kW}$, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is:

Open solution

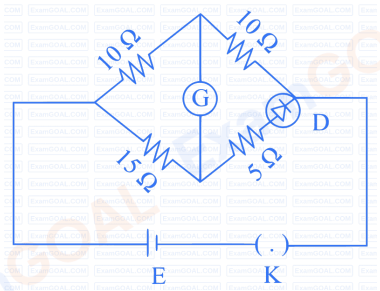
(A) $1:1$ (B) $2:9$ (C) $1:2$ (D) $2:3$

Q16. NEET 2024

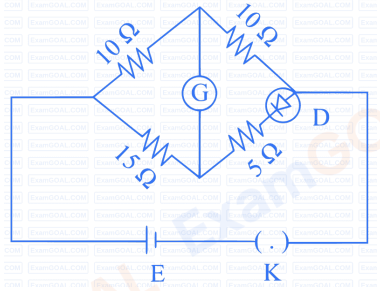
Choose the correct circuit which can achieve the bridge balance.

Open solution

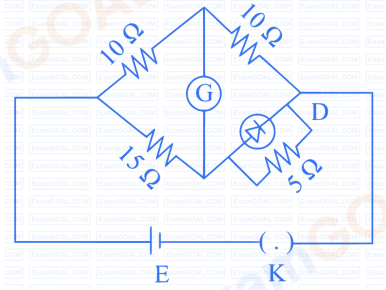
(A)



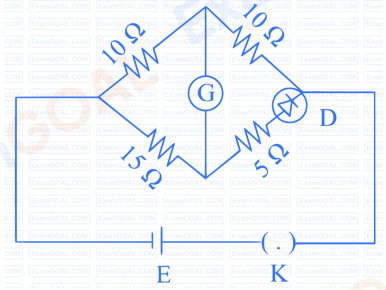
(B)



(C)



(D)



Q17. NEET 2023 Manipur

A certain wire A has resistance $81\,\Omega$. The resistance of another wire B of same material and equal length but of diameter thrice the diameter of A will be :

Open solution

(A) $81\,\Omega$ (B) $9\,\Omega$ (C) $729\,\Omega$ (D) $243\,\Omega$

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Q18. NEET 2023 Manipur

A copper wire of radius $1\ \text{mm}$ contains 10^{22} free electrons per cubic metre. The drift velocity for free electrons when $10\ \text{A}$ current flows through the wire will be (Given, charge on electron $= 1.6 \times 10^{-19}\text{C}$) :

Open solution

(A) $\frac{6.25\times10^{4}}{\pi}\text{ms}^{-1}$

(B) $\frac{6.25}{\pi} \times 10^{3}\ \text{ms}^{-1}$

(C) $\frac{6.25}{\pi}\text{ms}^{-1}$

(D) $\frac{6.25\times10^{5}}{\pi}\text{ms}^{-1}$

Q19. NEET 2023 Manipur

The emf of a cell having internal resistance 1Ω is balanced against a length of $330\ \text{cm}$ on a potentiometer wire. When an external resistance of 2Ω is connected across the cell, the balancing length will be :

Open solution

(A) 220 cm

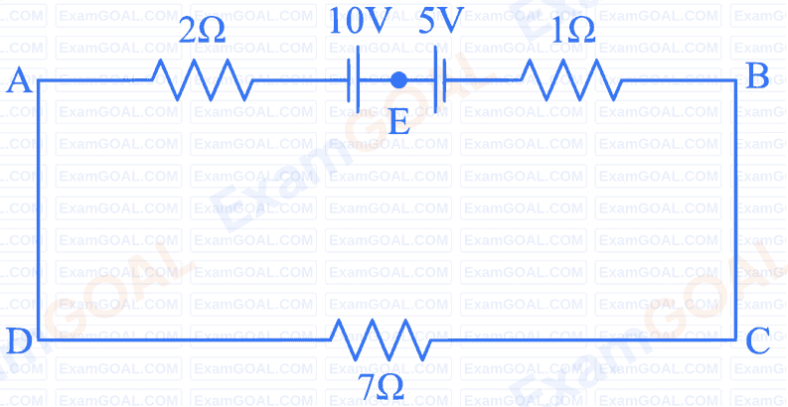
(B) 330 cm

(C) 115 cm

(D) 332 cm

Q20. NEET 2023

The magnitude and direction of the current in the following circuit is :-



Open solution

(A) 0.5 A from A to B through E

(B) $\frac{5}{9}A$ from A to B through E

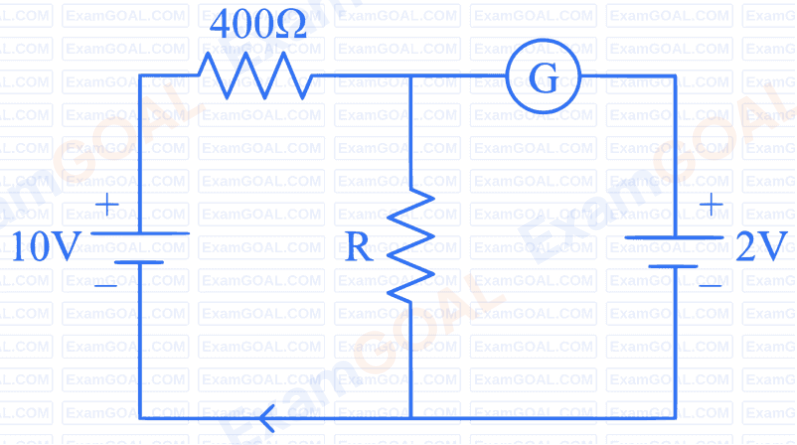
(C) 1.5 A from B to A through E

(D) 0.2 A from B to A through E

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Q21. NEET 2023

If the galvanometer G does not show any deflection in the circuit shown, the value of R is given by:



Open solution

(A) 50 Ω (B) 100 Ω (C) 400 Ω (D) 200 Ω

Q22. NEET 2023

Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%)\Omega$. The colour of third band must be :

(A) Green (B) Orange (C) Yellow (D) Red

Open solution

Q23. NEET 2023

The resistance of platinum wire at $0^{\circ}\mathrm{C}$ is $2\,\Omega$ and $6.8\,\Omega$ at $80^{\circ}\mathrm{C}$. The temperature coefficient of resistance of the wire is :

(A) $3 \times 10^{-3}\mathrm{C}^{-1}$ (B) $3 \times 10^{-2}\mathrm{C}^{-1}$

(C) $3 \times 10^{-10}\mathrm{C}^{-1}$ (D) $3 \times 10^{-4}\mathrm{C}^{-1}$

Open solution

Q24. NEET 2023

10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is :

(A) 100 (B) 1 (C) 1000 (D) 10

Open solution

Q25. NEET 2022 Phase 2

The reciprocal of resistance is :

(A) conductance (B) reactance (C) mobility (D) conductivity

Open solution

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Q26. NEET 2022 Phase 2

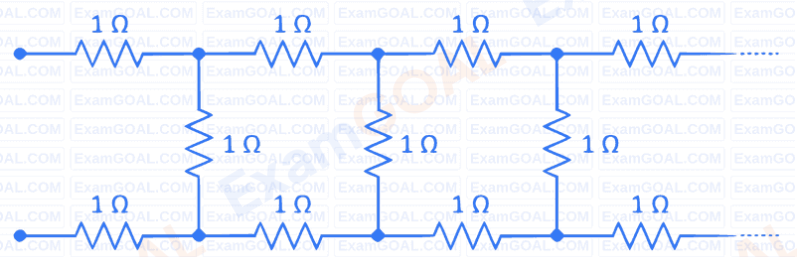
A cell of emf 4 V and internal resistance 0.5 Ω is connected to a 7.5 Ω external resistance. The terminal potential difference of the cell is

(A) 0.375 V (B) 3.75 V (C) 4.25 V (D) 4 V

Open solution

Q27. NEET 2022 Phase 2

The equivalent resistance of the infinite network given below is :



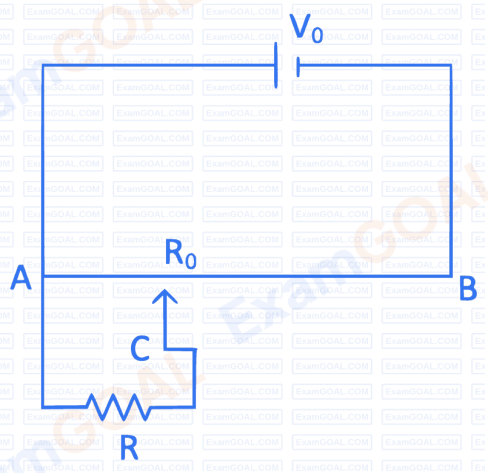
(A) $\left(1+\sqrt{5}\right)\Omega$ (B) $2\,\Omega$

(C) $\left(1+\sqrt{2}\right)\Omega$ (D) $\left(1+\sqrt{3}\right)\Omega$

Open solution

Q28. NEET 2022 Phase 2

The sliding contact C is at one fourth of the length of the potentiometer wire (AB) from A as shown in the circuit diagram. If the resistance of the wire AB is R_0, then the potential drop (V) across the resistor R is



(A) $\frac{2V_0R}{2R_0+3R}$ (B) $\frac{4V_0R}{3R_0+16R}$ (C) $\frac{4V_0R}{3R_0+R}$ (D) $\frac{2V_0R}{4R_0+R}$

Open solution

Q29. NEET 2022 Phase 1

Two resistors of resistance, 100 Ω and 200 Ω are connected in parallel in an electrical circuit. The ratio of the thermal energy developed in 100 Ω to that in 200 Ω in a given time is

(A) 1 : 2 (B) 2 : 1 (C) 1 : 4 (D) 4 : 1

Open solution

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Q30. NEET 2022 Phase 1

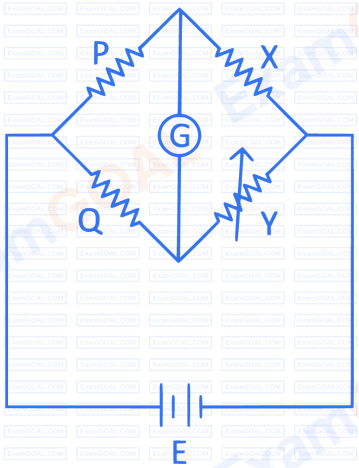
A copper wire of length 10 m and radius $\left(\frac{10^{-2}}{\sqrt{\pi}}\right)$ m has electrical resistance of 10 Ω. The current density in the wire for an electric field strength of 10 (V/m) is

(A) 10^4 A/m^2 (B) 10^6 A/m^2 (C) 10^{-5} A/m^2 (D) 10^5 A/m^2

Open solution

Q31. NEET 2022 Phase 1

A wheatstone bridge is used to determine the value of unknown resistance X by adjusting the variable resistance Y as shown in the figure. For the most precise measurement of X, the resistance P and Q



(A) Should be approximately equal to 2X

(B) Should be approximately equal and are small

(C) Should be very large and unequal

(D) Do not play any significant role

Open solution

Q32. NEET 2021

In a potentiometer circuit a cell of EMF 1.5 V gives balance point at 36 cm length of wire. If another cell of EMF 2.5 V replaces the first cell, then at what length of the wire, the balance point occurs?

(A) 62 cm (B) 60 cm (C) 21.6 cm (D) 64 cm

Open solution

Q33. NEET 2021

The effective resistance of a parallel connection that consists of four wires of equal length, equal area of cross-section and same material is 0.25Ω. What will be the effective resistance if they are connected in series?

(A) 4Ω (B) 0.25Ω (C) 0.5Ω (D) 1Ω

Open solution

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Q34. NEET 2021

Column - I gives certain physical terms associated with flow of current through a metallic conductor. Column-II gives some mathematical relations involving electrical quantities. Match column-I and column-II with appropriate relations.

Open solution

	Column - I		Column - II
(A)	Drift Velocity	(P)	$\frac{m}{ne^2\rho}$
(B)	Electrical Resistivity	(Q)	nev_d
(C)	Relaxation Period	(R)	$\frac{eE}{m}\tau$
(D)	Current Density	(S)	$\frac{E}{J}$

(A) (A)-(R); (B)-(Q); (C)-(S); (D)-(P)

(B) (A)-(R); (B)-(S); (C)-(P); (D)-(Q)

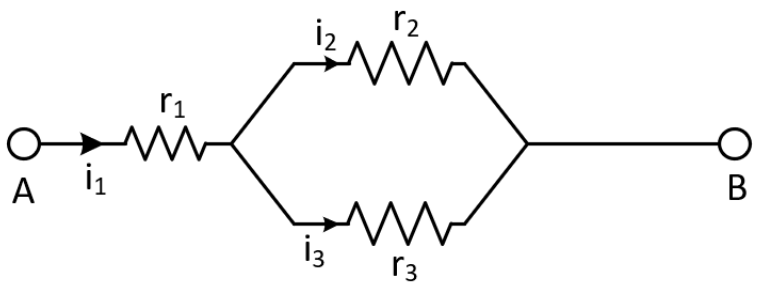
(C) (A)-(R); (B)-(S); (C)-(Q); (D)-(P)

(D) (A)-(R); (B)-(P); (C)-(S); (D)-(Q)

Q35. NEET 2021

Three resistors having resistances r_1, r_2 and r_3 are connected as shown in the given circuit. The ratio $\frac{i_3}{i_1}$ of currents in terms of resistances used in the circuit is :

Open solution



(A) $\frac{r_2}{r_1+r_3}$

(B) $\frac{r_1}{r_2+r_3}$

(C) $\frac{r_2}{r_2+r_3}$

(D) $\frac{r_1}{r_1+r_2}$

Q36. NEET 2020 Phase 1

A resistance wire connected in the left gap of a metre bridge balances a 10 Ω resistance in the right gap at a point which divides the bridge wire in the ratio 3:2. If the length of the resistance wire is 1.5 m, then the length of 1 Ω of the resistance wire is :

Open solution

(A) $1.0 \times 10^{-1} m$

(B) $1.5 \times 10^{-1} m$

(C) $1.5 \times 10^{-2} m$

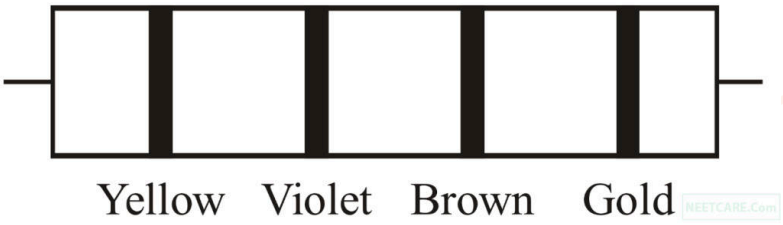
(D) $1.0 \times 10^{-2} m$

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Q37. NEET 2020 Phase 1

Open solution

The color code of a resistance is given below :



The values of resistance and tolerance, respectively, are

(A) 47 kΩ , 10% (B) 4.7 kΩ , 5% (C) 470 Ω , 5% (D) 470 kΩ , 5%

Q38. NEET 2020 Phase 1

Open solution

A charged particle having drift velocity 7.5×10^{-4} ms^{-1} in an electric field of 3×10^{-10} Vm^{-1} has a mobility in m^2 $V^{-1}s^{-1}$ of :

(A) 2.5×10^{6} (B) 2.5×10^{-6} (C) 2.25×10^{-15} (D) 2.25×10^{15}

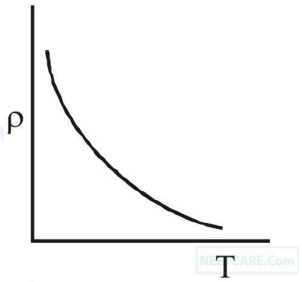
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Q39. NEET 2020 Phase 1

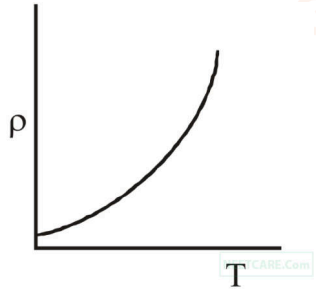
Which of the following graph represents the variation of resistivity (ρ) with temperature (T) for copper?

Open solution

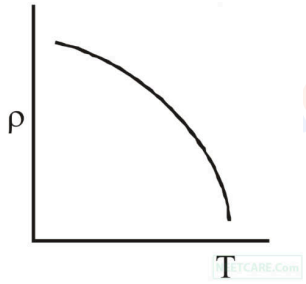
(A)



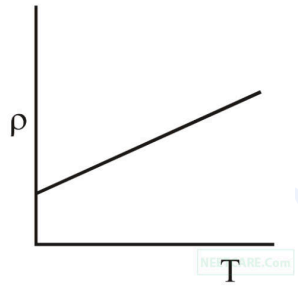
(B)



(C)



(D)



Q40. NEET 2020 Phase 1

The solids which have the negative temperature coefficient of resistance are :

(A) insulators only

(B) semiconductors only

(C) insulators and semiconductors

(D) metals

Open solution

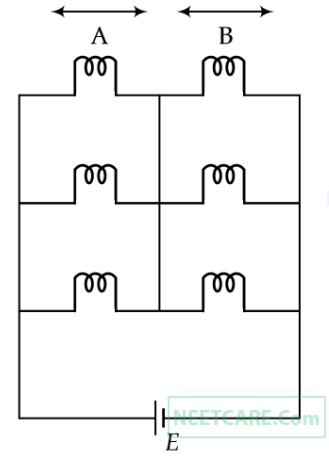
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Q41. NEET 2019

Six similar bulbs are connected as shown in the figure with a DC source of emf E, and zero internal resistance.

Open solution

The ratio of power consumption by the bulbs when (i) all are glowing and (ii) in the situation when two from section A and one from section B are glowing, will be :

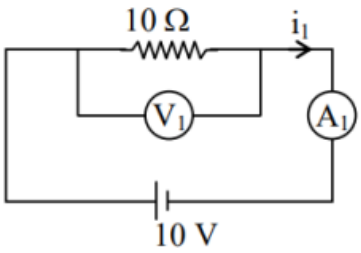


(A) 4 : 9 (B) 9 : 4 (C) 1 : 2 (D) 2 : 1

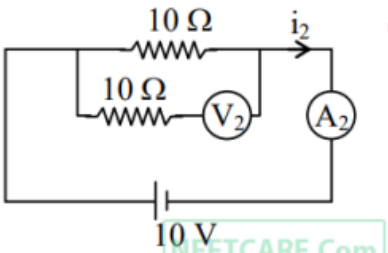
Q42. NEET 2019

In the circuits shown below, the readings of voltmeters and the ammeters will be :

Open solution



Circuit 1



Circuit 2

(A) $V_1 = V_2$ and $i_1 = i_2$ (B) $V_1 > V_2$ and $i_1 > i_2$

(C) $V_1 > V_2$ and $i_1 = i_2$ (D) $V_1 = V_2$ and $i_1 > i_2$

Q43. NEET 2019

Which of the following acts as a circuit protection device?

Open solution

(A) inductor (B) conductor (C) switch (D) fuse

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Q44. NEET 2018

A set of n equal resistors, of value R each, are connected in series to a battery of emf E and internal resistance R. The current drawn is I. Now, the n resistors are connected in parallel to the same battery. Then the current drawn from battery becomes 10I. The value of n is

Open solution

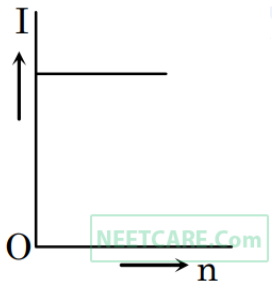
(A) 10 (B) 11 (C) 20 (D) 9

Q45. NEET 2018

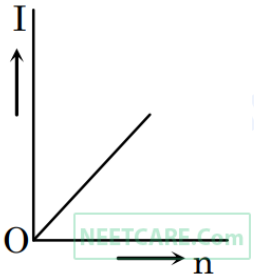
A battery consists of a variable number n of identical cells (having internal resistance r each) which are connected in series. The terminals of the battery are short-circuited and the current I is measured. Which of the graphs shows the correct relationship between I and n?

Open solution

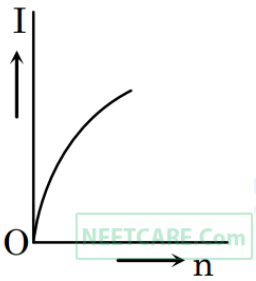
(A)



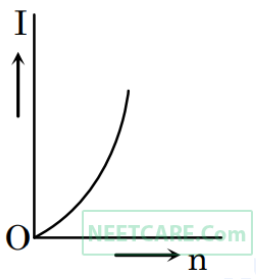
(B)



(C)



(D)



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Q46. NEET 2018

A carbon resistor of $(47 \pm 4.7)\ k\Omega$ is to be marked with rings of different colours for its identification. The colour code sequence will be

(A) Violet – Yellow – Orange – Silver
(B) Yellow – Violet – Orange – Silver
(C) Yellow – Green – Violet – Gold
(D) Green – Orange – Violet – Gold

Open solution

Q47. NEET 2017

The resistance of a wire is 'R' ohm. If it is melted and stretched to 'n' times its original length. its new resistance will be

(A) $\frac{R}{n}$ (B) n^2R (C) $\frac{R}{n^2}$ (D) nR

Open solution

Q48. NEET 2017

A potentiometer is an accurate and versatile device to make electrical measurements of EMF because the method involves

(A) potential gradients
(B) a condition of no current flow through the galvanometer
(C) a combination of cells, galvanometer and resistances
(D) cells

Open solution

Q49. NEET 2016 Phase 2

The potential difference $(V_A - V_B)$ between the points A and B in the given figure is



(A) − 3V (B) +3 V (C) +6 V (D) +9 V

Open solution

Q50. NEET 2016 Phase 2

A filament bulb (500 W, 100 V) is to be used in a 230 V main supply. When a resistance R is connected in series, it works perfectly and the bulb consumes 500 W. The value of R is

(A) 230 Ω (B) 46 Ω (C) 26 Ω (D) 13 Ω

Open solution

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Q51. NEET 2016 Phase 1

A potentiometer wire is 100 cm long and a constant potential difference is maintained across it. Two cells are connected in series first to support one another and then in opposite direction. The balance points are obtained at 50 cm and 10 cm from the positive end of the wire in the two cases. The ratio of emf's is

Open solution

(A) 3 : 4 (B) 3 : 2 (C) 5 : 1 (D) 5 : 4

Q52. NEET 2016 Phase 1

The charge flowing through a resistance R varies with time t as Q = at $-$ bt^2, where a and b are positive constants. The total heat produced in R is

Open solution

(A) $\frac{a^3R}{2b}$ (B) $\frac{a^3R}{b}$ (C) $\frac{a^3R}{6b}$ (D) $\frac{a^3R}{3b}$

Q53. AIPMT 2015

Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are their conductivities, the effective conductivity of the combination is

Open solution

(A) $\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$ (B) $\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

(C) $\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$ (D) $\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$

Q54. AIPMT 2015

A circuit contains an ammeter, a battery of 30 V and a resistance 40.8 ohm all connected in series. If the ammeter has a coil of resistance 480 ohm and a shunt of 20 ohm, the reading in the ammeter will be

Open solution

(A) 2A (B) 1A (C) 0.5 A (D) 0.25 A

Q55. AIPMT 2015

A potentiometer wire of length L and a resistance r are connected in series with a battery of e.m.f. E_0 and a resistance r_1. An unknown e.m.f. E is balanced at a length l of the potentiometer wire. The e.m.f. E will be given by

Open solution

(A) $\frac{E_0 l}{L}$ (B) $\frac{LE_0 r}{(r+r_1)l}$

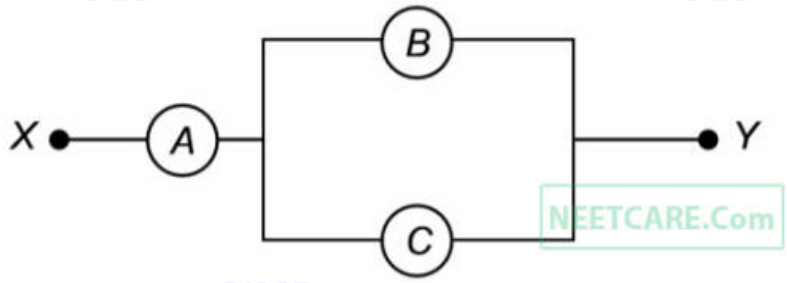
(C) $\frac{LE_0 r}{lr_1}$ (D) $\frac{E_0 r}{(r+r_1)} \cdot \frac{l}{L}$

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Q56. AIPMT 2015 Cancelled Paper

A, B and C are voltmeters of resistance R, 1.5 R and 3R respectively as shown in the figure. When some potential difference is applied between X and Y, the voltmeter readings are V_A, V_B and V_C respectively, Then

Open solution



(A) $V_A = V_B \neq V_C$

(B) $V_A \neq V_B \neq V_C$

(C) $V_A = V_B = V_C$

(D) $V_A \neq V_B = V_C$

Q57. AIPMT 2015 Cancelled Paper

Across a metallic conductor of non-uniform cross section a constant potential difference is applied. The quantity which remains constant along the conductor is

(A) drift velocity (B) electric field (C) current density (D) current

Open solution

Q58. AIPMT 2015 Cancelled Paper

A potentiometer wire has length 4 m and resistance 8 Ω. The resistance that must be connected in series with the wire and an accumulator of e.m.f. 2 V, so as to get a potential gradient 1 mV per cm on the wire is

(A) 44 Ω (B) 48 Ω (C) 32 Ω (D) 40 Ω

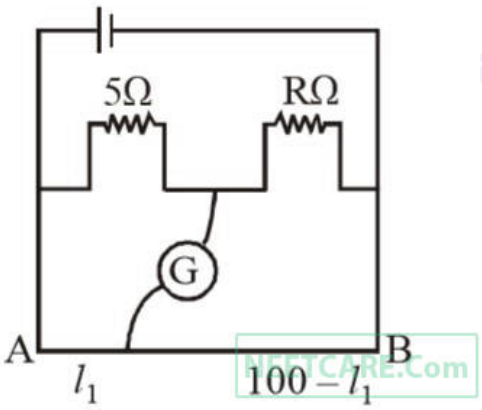
Open solution

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Q59. AIPMT 2014

The resistances in the two arms of the meter bridge are 5 Ω and RΩ respectively. When the resistance R is shunted with an equal resistance, the new balance point is at $1.6l_1$. The resistance R is

Open solution



(A) 10 Ω (B) 15 Ω (C) 20 Ω (D) 25 Ω

Q60. AIPMT 2014

Two cities are 150 km apart. Electric power is sent from one city to another city through copper wires. The fall of potential per km is 8 volt and the average resistance per km is 0.5 Ω. The power loss in the wire is

Open solution

(A) 19.2 W (B) 19.2 kW (C) 19.2 J (D) 12.2 kW

Q61. AIPMT 2014

A potentiometer circuit has been set up for finding the internal resistance of a given cell. The main battery, used across the potentiometer wire, has an emf of 2.0 V and a negligible internal resistance. The potentiometer wire itself is 4 m long. When the resistance R, connected across the given cell, has values of (i) infinity, (ii) 9.5 Ω
the balancing lengths on the potentiometer wire are found to be 3 m and 2.85 m, respectively. The value of internal resistance of the cell is

Open solution

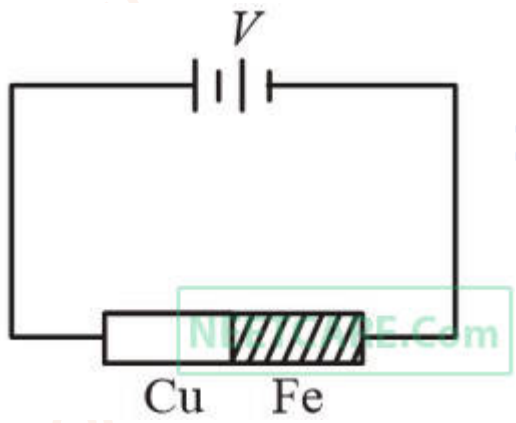
(A) 0.25 Ω (B) 0.95 Ω (C) 0.5 Ω (D) 0.75 Ω

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Q62. NEET 2013 (Karnataka)

Two rods are joined end to end, as shown. Both have a cross-sectional area of 0.01 cm^2. Each is 1 meter long. One rod is of copper with a resistivity of 1.7×10^{-6} ohm-centimeter, the other is of iron with a resistivity of 10^{-5} ohm-centimeter.

Open solution



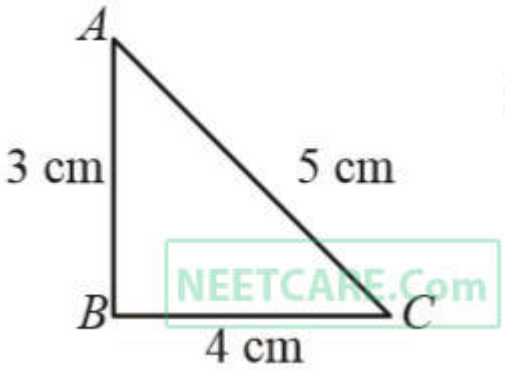
How much voltage is required to produce a current of 1 ampere in the rods?

(A) 0.00145 V (B) 0.0145 V (C) 1.7×10^{-6} V (D) 0.117 V

Q63. NEET 2013 (Karnataka)

A 12 cm wire is given a shape of a right angled triangle ABC having sides 3 cm, 4 cm and 5 cm as shown in the figure. The resistance between two ends (AB, BC, CA) of the respective sides are measured one by one. Their ratio is

Open solution



(A) 9 : 16 : 25 (B) 27 : 32 : 35 (C) 21 : 24 : 25 (D) 3 : 4 : 5

Q64. NEET 2013 (Karnataka)

Ten identical cells connected in series are needed to heat a wire of length one meter and radius 'r' by 10^oC in time 't'. How many cells will be required to heat the wire of length two meter of the same radius by the same temperature in time 't' ?

Open solution

(A) 20 (B) 30 (C) 40 (D) 10

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Q65. NEET 2013

The resistances of the four arms P, Q, R and S in a Wheatstone's bridge are 10 ohm, 30 ohm, 30 ohm and 90 ohm, respectively. The e.m.f. and internal resistance of the cell are 7 volt and 5 ohm respectively. If the galvanometer resistance is 50 ohm, the current drawn from the cell will be

(A) 0.1 A (B) 2.0 A (C) 1.0 A (D) 0.2 A

Open solution

Q66. NEET 2013

The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10 Ω is

(A) 0.8 Ω (B) 1.0 Ω (C) 0.2 Ω (D) 0.5 Ω

Open solution

Q67. NEET 2013

A wire of resistance 4 Ω is stretched to twice its original length. The resistance of stretched wire would be

(A) 8 Ω (B) 16 Ω (C) 2 Ω (D) 4 Ω

Open solution

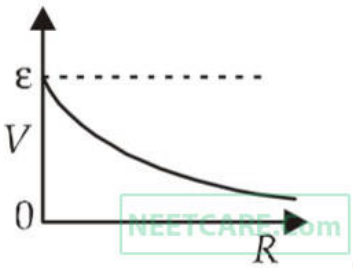
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Q68. AIPMT 2012 Mains

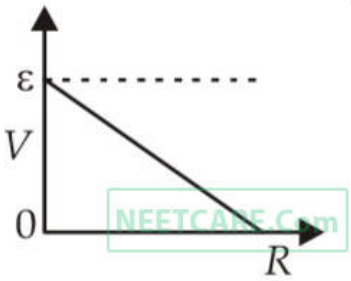
A cell having an emf ε and internal resistance r is connected across a variable external resistance R. As the resistance R is increased, the plot of potential difference V across R is given by

Open solution

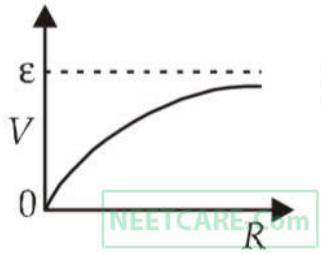
(A)



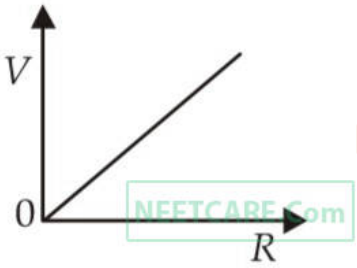
(B)



(C)



(D)

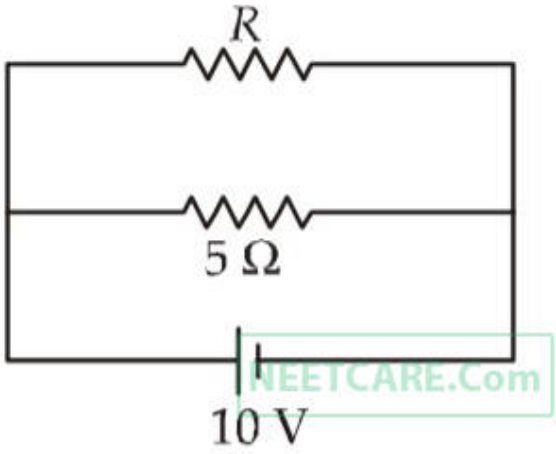


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Q69. AIPMT 2012 Mains

The power dissipated in the circuit shown in the figure is 30 watts. The value of R is

Open solution



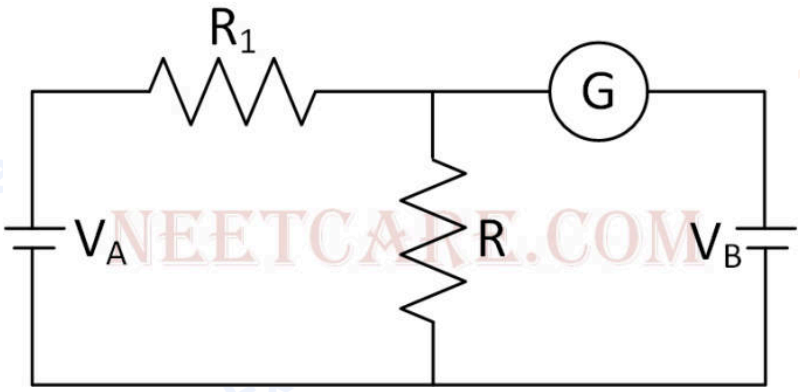
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(A) 20 Ω (B) 15 Ω (C) 10 Ω (D) 30 Ω

Q70. AIPMT 2012 Prelims

In the circuit shown the cells A and B have negligible resistances. For V_A = 12 V, R_1 = 500 Ω and R = 100 Ω the galvanometer (G) shows no deflection. The value of V_B is

Open solution



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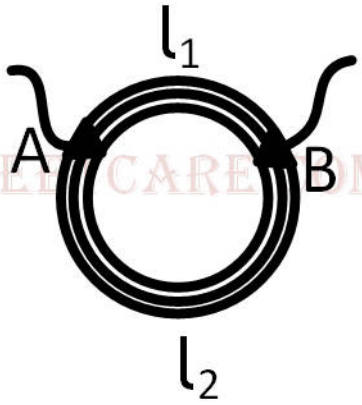
(A) 4 V (B) 2 V (C) 12 V (D) 6 V

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Q71. AIPMT 2012 Prelims

A ring is made of a wire having a resistance $R_0 = 12\,\Omega$. Find the points A and B, as shown in the figure, at which a current carrying conductor should be connected so that the resistance R of the sub circuit between these point is equal to $\frac{8}{3}\Omega$.

Open solution



(A) $\frac{l_1}{l_2} = \frac{5}{8}$
(B) $\frac{l_1}{l_2} = \frac{1}{3}$
(C) $\frac{l_1}{l_2} = \frac{3}{8}$
(D) $\frac{l_1}{l_2} = \frac{1}{2}$

Q72. AIPMT 2012 Prelims

If voltage across a bulb rated 220 volt-100 watt drops by 2.5% of its rated value, the percentage of the rated value by which the power would decrease is

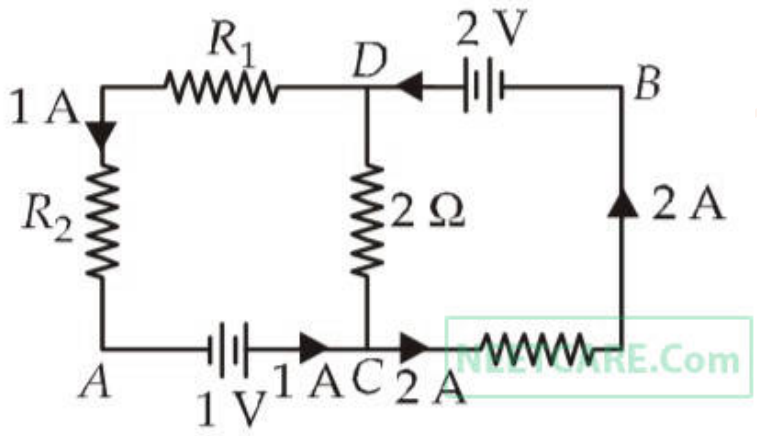
Open solution

(A) 20%
(B) 2.5%
(C) 5%
(D) 10%

Q73. AIPMT 2011 Mains

In the circuit shown in the figure, if the potential at point A is taken to be zero, the potential at point B is

Open solution



(A) +1 V
(B) −1 V
(C) +2 V
(D) −2 V

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Q74. AIPMT 2011 Mains

A thermocouple of negligible resistance produces an e.m.f. of 40 μV/°C in the linear range of temperature. A galvanometer of resistance 10 ohm whose sensitivity is 1 μA/division, is employed with the thermocouple. The smallest value of temperature difference that can be detected by the system will be

Open solution

(A) 1°C (B) 0.5 °C (C) 0.1°C (D) 0.25°C

Q75. AIPMT 2011 Prelims

A current of 2 A flows through a 2 Ω resistor when connected across a battery. The same battery supplies a current of 0.5 A when connected across a 9 Ω resistor. The internal resistance of the battery is

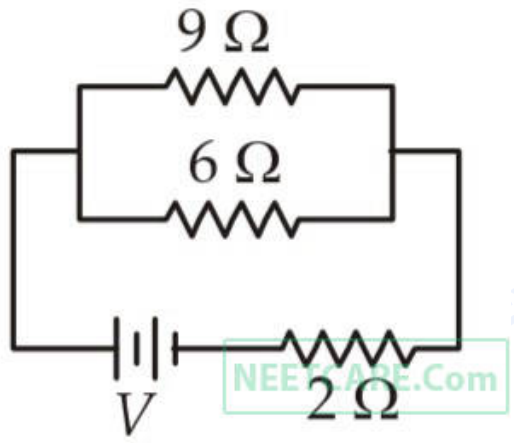
Open solution

(A) 0.5 Ω (B) 1/3 Ω (C) 1/4 Ω (D) 1 Ω

Q76. AIPMT 2011 Prelims

If power dissipated in the 9 Ω resistor in the circuit shown is 36 watt, the potential difference across the 2 Ω resistor is

Open solution



(A) 4 volt (B) 8 volt (C) 10 volt (D) 2 volt

Q77. AIPMT 2010 Prelims

Consider the following two statements.
(A) Kirchoff's junction law follows from the conservation of charge.
(B) Kirchhoff's loop law follows from the conservation of energy.

Which of the following is correct?

Open solution

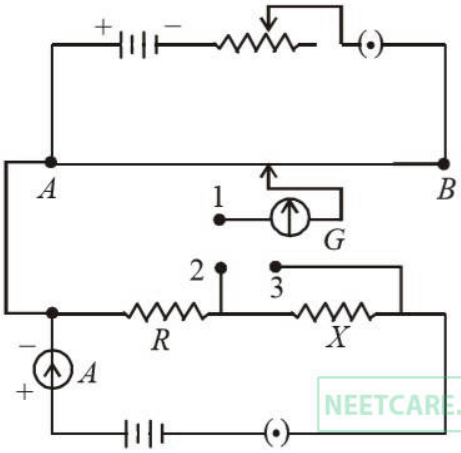
(A) Both (A) and (B) are wrong (B) (A) is correct and (B) is wrong
(C) (A) is wrong and (B) is correct (D) Both (A) and (B) are correct

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Q78. AIPMT 2010 Prelims

Open solution

A potentiometer circuit is set up as shown. The potential gradient, across the potentiometer wire, is k volt/cm and the ammeter, present in the circuit, reads 1.0 A when two way key is switched off. The balance points, when the key between the terminals (i) 1 and 2 (ii) 1 and 3, is plugged in, are found to be at length l_1 cm and l_2 cm respectively. The magnitudes, of the resistors R and X, in ohms, are then, equal, respectively, to



(A) $k(l_2 - l_1)$ and kl_2

(B) kl_1 and $k(l_2 - l_1)$

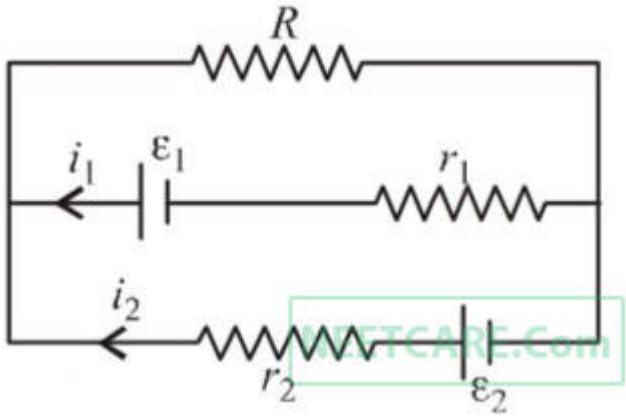
(C) $k\,(l_2 - l_1)$ and kl_1

(D) kl_1 and kl_2

Q79. AIPMT 2009

Open solution

See the electrical circuit shown in this figure. Which of the following equations is a correct equation for it ?



(A) $\varepsilon_2 - i_2 r_2 - \varepsilon_1 - i_1 r_1 = 0$

(B) $-\varepsilon_2 - (i_1 + i_2)R + i_2 r_2 = 0$

(C) $\varepsilon_1 - (i_1 + i_2)R + i_1 r_1 = 0$

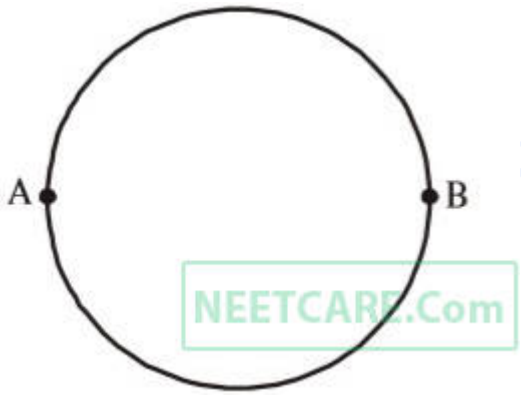
(D) $\varepsilon_1 - (i_1 + i_2)R - i_1 r_1 = 0$

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Q80. AIPMT 2009

A wire of resistance 12 ohms per meter is bent to form a complete circle of radius 10 cm. The resistance between its two diametrically opposite points, A and B as shown in the figure is

Open solution



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(A) $3\,\Omega$ (B) $6\pi\,\Omega$ (C) $6\,\Omega$ (D) $0.6\pi\,\Omega$

Q81. AIPMT 2009

A student measures the terminal potential difference (V) of a cell (of emf ε and internal resistance r) as a function of the current (I) flowing through it. The slope, and intercept, of the graph between V and I, then respectively, equal

Open solution

(A) $-$ r and ε (B) r and $-\varepsilon$ (C) $-\varepsilon$ and r (D) ε and $-$ r

Q82. AIPMT 2009

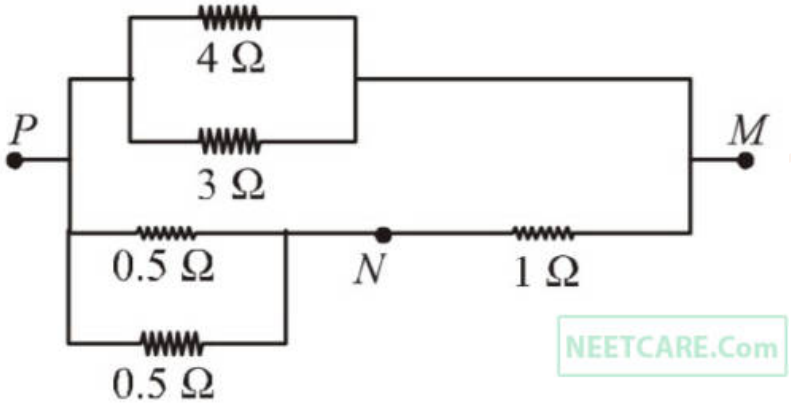
The mean free path of electrons in a metal is 4×10^{-8} m. The electric field which can give on an average 2 eV energy to an electron in the metal will be in units V/m

Open solution

(A) 5×10^{-11} (B) 8×10^{-11} (C) 5×10^{7} (D) 8×10^{7}

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Q83. AIPMT 2008



Open solution

In the circuit shown, the current through the 4 Ω resistor is 1 amp when the points P and M are connected to a d.c. voltage source. The potential difference between the points M and N is

(A) 0.5 volt (B) 3.2 volt (C) 1.5 volt (D) 1.0 volt

Q84. AIPMT 2008

A wire of a certain material is stretched slowly by ten percent. Its new resistance and specific resistance become respectively

(A) both remain the same (B) 1.1 times, 1.1 times
(C) 1.2 times, 1.1 times (D) 1.21 times, same

Open solution

Q85. AIPMT 2008

A cell can be balanced against 110 cm and 100 cm of potentiometer wire, respectively with and without being short circuited through a resistance of 10 Ω. Its internal resistance is

(A) 2.0 ohm (B) zero (C) 1.0 ohm (D) 0.5 ohm

Open solution

Q86. AIPMT 2008

An electric kettle takes 4 A current at 220 V. How much time will it take to boil 1 kg of water from temperature 20°C ? The temperature of boiling water is 100°C

(A) 12.6 min (B) 4.2 min (C) 6.3 min (D) 8.4 min

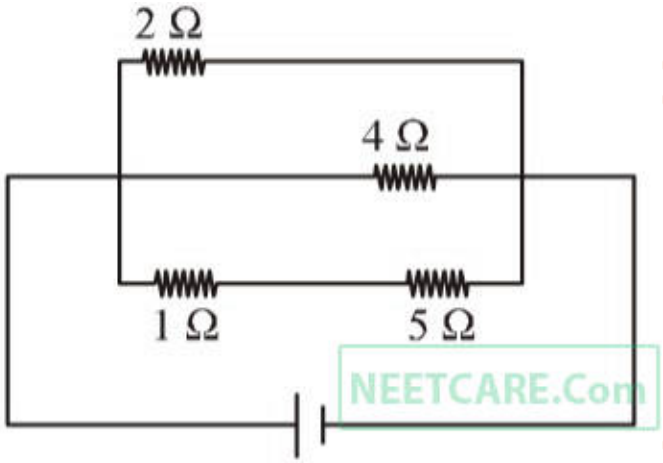
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Q87. AIPMT 2008

A current of 3 amp. flows through the 2 Ω resistor shown in the circuit. The power dissipated in the 5 Ω resistor is

Open solution



(A) 1 watt (B) 5 watt (C) 4 watt (D) 2 watt

Q88. AIPMT 2007

Three resistances, P, Q, R each of 2Ω and an unknown resistance S from the four arms of a Wheatstone bridge circuit. When a resistance of 6 Ω is connected in parallel to S the bridge gets balanced. What is the value of S?

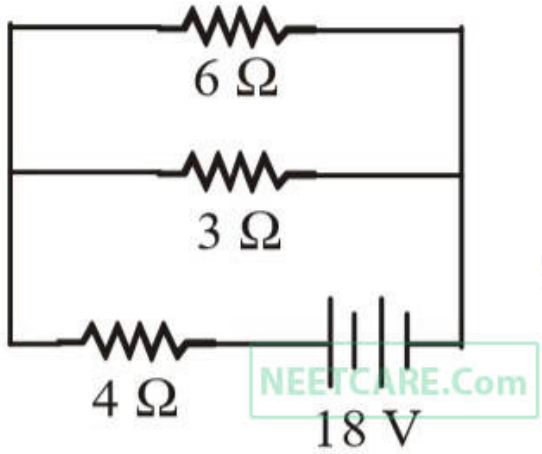
Open solution

(A) 3 Ω (B) 6 Ω (C) 1 Ω (D) 2 Ω.

Q89. AIPMT 2007

The total power dissipated in watt in the circuit shown here is

Open solution



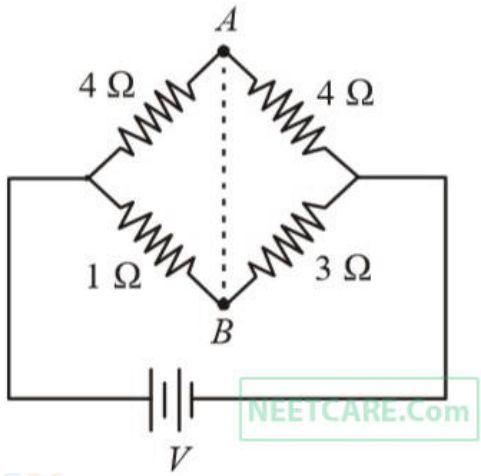
(A) 40 (B) 54 (C) 4 (D) 16

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Q90. AIPMT 2006

In the circuit shown, if a conducting wire is connected between points A and B, the current in this wire will

Open solution



(A) flow from B to A

(B) flow from A to B

(C) flow in the direction which will be decided by the value of V be zero.

(D) be zero

Q91. AIPMT 2006

Kirchhoff's first and second laws of electrical circuits are consequences of

Open solution

(A) conservation of energy and electric charge respectively

(B) conservation of energy

(C) conservation of electric charge and energy respectively

(D) conservation of electric charge.

Q92. AIPMT 2006

Two cells, having the same e.m.f. are connected in series through an external resistance R. Cells have internal resistances r_1 and r_2 ($r_1 > r_2$) respectively. When the circuit is closed, the potential difference across the first cell is zero. The value of R is

Open solution

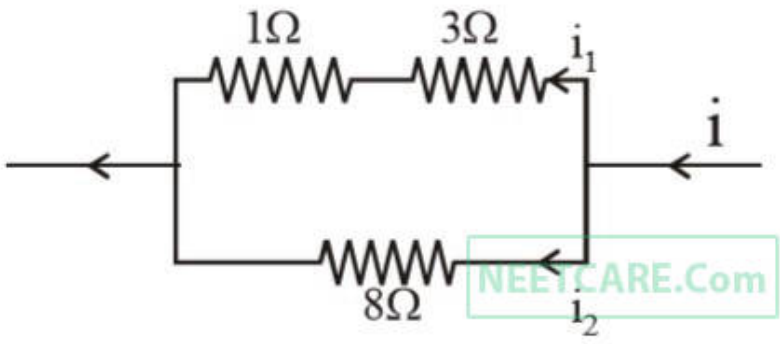
(A) $r_1 + r_2$ (B) $r_1 - r_2$ (C) $\frac{r_1+r_2}{2}$ (D) $\frac{r_1-r_2}{2}$

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Q93. AIPMT 2006

Power dissipated across the 8 Ω resistor in the circuit shown here is 2 watt. The power dissipated in watt units across the 3 Ω resistor is

Open solution



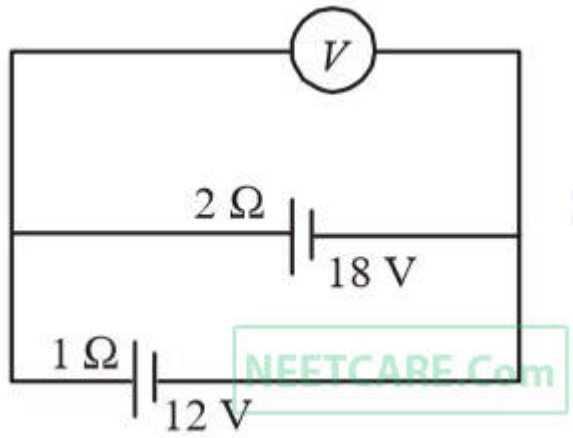
NEETCARE.Com

(A) 3.0 (B) 2.0 (C) 1.0 (D) 0.5

Q94. AIPMT 2005

Two batteries, one of emf 18 volts and internal resistance 2 Ω and the other of emf 12 volts and internal resistance 1 Ω, are connected as shown. The voltmeter V will record a reading of

Open solution



NEETCARE.Com

(A) 30 volt (B) 18 volt (C) 15 volt (D) 14 volt

Q95. AIPMT 2005

When a wire of uniform cross-section a length l and resistance R is bent into a complete circle. resistance between any two of diametrically opposite points will be

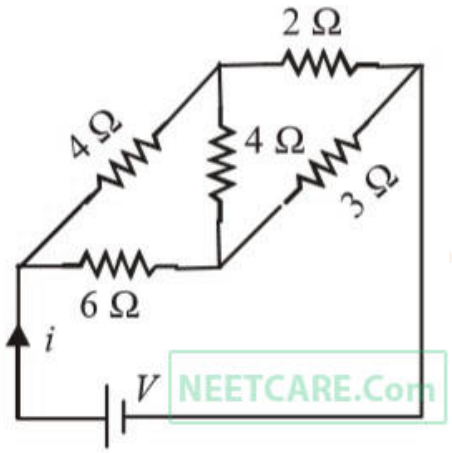
Open solution

(A) R/4 (B) 4R (C) R/8 (D) R/2

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Q96. AIPMT 2005

For the network shown in the figure the value of the current i is



Open solution

(A) $\frac{9V}{35}$ (B) $\frac{18V}{5}$ (C) $\frac{5V}{9}$ (D) $\frac{5V}{18}$

Q97. AIPMT 2005

A 5-ampere fuse wire can withstand a maximum power of 1 watt in the circuit. The resistance of the fuse wire is

(A) 0.04 ohm (B) 0.2 ohm (C) 5 ohm (D) 0.4 ohm

Open solution

Q98. AIPMT 2004

The electric resistance of a certain wire of iron is R. If its length and radius are both doubled, then

(A) The resistance will be doubled and the specific resistance will be halved.

(B) The resistance will be halved and the specific resistance will remain unchanged.

(C) The resistance will be halved and the specific resistance will be doubled.

(D) The resistance and the specific resistance, will both remain unchanged.

Open solution

Q99. AIPMT 2004

Resistance n, each of r ohm, when connected in parallel give an equivalent resistance of R ohm. If these resistances were connected in series, the combination would have a resistance in ohms, equal to

(A) n^2R (B) R/n^2 (C) R/n (D) nR

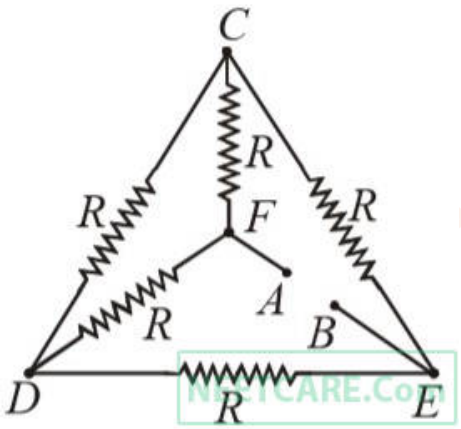
Open solution

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Q100. AIPMT 2004

Five equal resistances each of resistances R are connected as shown in the figure. A battery of V volts is connected between A and B. The current flowing in AFCEB will be

Open solution



(A) $\frac{3V}{R}$ (B) $\frac{V}{R}$ (C) $\frac{V}{2R}$ (D) $\frac{2V}{R}$

Q101. AIPMT 2004

A 6 volt battery is connected to the terminals of a three metre long wire of uniform thickness and resistance of 100 ohm. The difference of potential between two points on the wire separated by a distance of 50 cm will be

Open solution

(A) 2 volt (B) 3 volt (C) 1 volt (D) 1.5 volt

Q102. AIPMT 2004

when three identical bulbs of 60 watt, 200 volt rating are connected in series to a 200 volt supply, the power drawn by them will be

Open solution

(A) 60 watt (B) 180 watt (C) 10 watt (D) 20 watt

Q103. AIPMT 2004

In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60 W bulb for use in India is R, the resistance of a 60 W bulb for use in USA will be

Open solution

(A) R (B) 2R (C) R/4 (D) R/2

Q104. AIPMT 2003

In a Wheatstone's bridge all the four arms have equal resistance R. If the resistance of the galvanometer arm is also R, the equivalent resistance of the combination as seen by the battery is

Open solution

(A) R/4 (B) R/2 (C) R (D) 2R

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Q105. AIPMT 2003

Two 220 volt, 100 watt bulbs are connected first in series and then in parallel. Each time the combination is connected to a 220 volt a.c. supply line. The power drawn by the combination in each case respectively will be

(A) 50 watt, 100 watt
(B) 100 watt, 50 watt
(C) 200 watt, 150 watt
(D) 50 watt, 200 watt

Open solution

Q106. AIPMT 2003

An electric kettle has two heating coils. When one of the coils is connected to an a.c. source, the water in the kettle boils in 10 minutes. When the other coil is used the water boils in 40 minutes. If both the coils are connected in parallel, the time taken by the same quantity of water to boil will be

(A) 8 minutes
(B) 4 minutes
(C) 25 minutes
(D) 15 minutes

Open solution

Q107. AIPMT 2003

Fuse wire is a wire of

(A) high resistance and high melting point
(B) high resistance and low melting point
(C) low resistance and low melting point
(D) low resistance and high melting point

Open solution

Q108. AIPMT 2002

For a cell terminal potential difference is 2.2V when circuit is open and reduce to 1.8 V when cell is connected to a resistance of R = 5 Ω. Determine internal resistance of cell (r)

(A) $\frac{10}{9}\Omega$
(B) $\frac{9}{10}\Omega$
(C) $\frac{11}{9}\Omega$
(D) $\frac{5}{9}\Omega$

Open solution

Q109. AIPMT 2002

Specific resistance of a conductor increases with

(A) increase in temperature
(B) increase in cross-section area
(C) increase in cross-section and decrease in length
(D) decrease in cross secton area.

Open solution

Q110. AIPMT 2001

Copper and silicon is cooled from 300 K to 60 K , the specific resistance

(A) decrease in copper but increase in silicon
(B) increase in copper but decrease in silicon
(C) increase in both
(D) decrease in both

Open solution

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Q111. AIPMT 2001

The resistance of each arm of the Wheatstone's bridge is 10 ohm. A resistance of 10 ohm is connected in series with a galvanometer then the equivalent resistance across the battery will be

(A) 10 ohm (B) 15 ohm (C) 20 ohm (D) 40 ohm

Open solution

Q112. AIPMT 2001

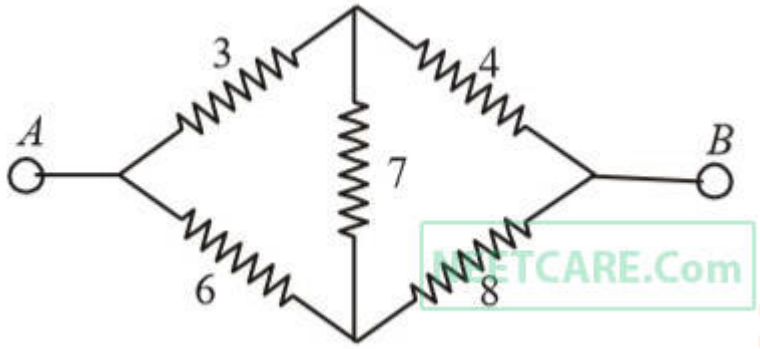
If specific resistance of a potentiometer wire is $10^{-7}\Omega$ m and current flow through it is 0.1 amp., cross-sectional area of wire is 10^{-6} m^2 then potential gradient will be

(A) 10^{-2} volt/m (B) 10^{-4} volt/m (C) 10^{-6} volt/m (D) 10^{-8} volt/m.

Open solution

Q113. AIPMT 2000

The net resistance of the circuit between A and B is



(A) $\frac{8}{3}\Omega$ (B) $\frac{14}{3}\Omega$ (C) $\frac{16}{3}\Omega$ (D) $\frac{22}{3}\Omega$

Open solution

Q114. AIPMT 2000

A car battery of emf 12 V and internal resistance $5 \times 10^{-2}\ \Omega$., receives a current of 60 amp, from external source, then terminal potential difference of battery is

(A) 12 V (B) 9 V (C) 15 V (D) 20 V

Open solution

Q115. AIPMT 2000

Two bulbs of (40 W, 200 V), and (100 W, 200 V). Then correct relation for their resistances

(A) $R_{40} < R_{100}$ (B) $R_{40} > R_{100}$

(C) $R_{40} = R_{100}$ (D) no relation can be predicted.

Open solution

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Q116. AIPMT 2000

The potentiometer is best for measuring voltage, as

(A) it has a sensitive galvanometer and gives null deflection

(B) it has wire of high resistance

(C) it measures p.d. in closed circuit

(D) it measures p.d in open circuit.

Open solution

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Answer Key

1. **B**	2. **D**	3. **A**	4. **C**	5. **A**
6. **D**	7. **D**	8. **A**	9. **D**	10. **B**
11. **A**	12. **D**	13. **C**	14. **B**	15. **B**
16. **A**	17. **B**	18. **B**	19. **A**	20. **A**
21. **B**	22. **B**	23. **B**	24. **A**	25. **A**
26. **B**	27. **D**	28. **B**	29. **B**	30. **D**
31. **B**	32. **B**	33. **A**	34. **B**	35. **C**
36. **A**	37. **C**	38. **A**	39. **B**	40. **C**
41. **B**	42. **A**	43. **D**	44. **A**	45. **A**
46. **B**	47. **B**	48. **B**	49. **D**	50. **C**
51. **B**	52. **C**	53. **C**	54. **C**	55. **D**
56. **C**	57. **D**	58. **C**	59. **B**	60. **B**
61. **C**	62. **D**	63. **B**	64. **A**	65. **D**
66. **D**	67. **B**	68. **C**	69. **C**	70. **B**
71. **D**	72. **C**	73. **A**	74. **D**	75. **B**
76. **C**	77. **D**	78. **B**	79. **D**	80. **D**
81. **A**	82. **C**	83. **B**	84. **D**	85. **C**
86. **C**	87. **B**	88. **A**	89. **B**	90. **A**
91. **C**	92. **B**	93. **A**	94. **D**	95. **A**
96. **D**	97. **A**	98. **B**	99. **A**	100. **C**
101. **C**	102. **D**	103. **C**	104. **C**	105. **D**
106. **A**	107. **B**	108. **A**	109. **A**	110. **A**
111. **A**	112. **A**	113. **B**	114. **C**	115. **B**
116. **A**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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